## TASK 2: Variables, Data Types & Console Input Application

### Tools:

- Primary: IntelliJ IDEA / Eclipse
- Alternatives: VS Code, Terminal + Text Editor

### Hints / Mini Guide:

1. Create a Java program that declares variables of all primitive data types and explain their memory usage.
2. Use the Scanner class to accept multiple types of user input from the console.
3. Implement basic arithmetic operations using user-provided input.
4. Demonstrate type casting by converting between compatible and incompatible data types.
5. Handle invalid input gracefully using conditional checks.
6. Print formatted output using System.out.printf.
7. Show difference between local, instance, and static variables.
8. Add inline comments explaining why specific data types were chosen.

### Deliverables:

- Interactive console-based calculator
- Clean, well-commented source code

### Interview Questions Related To Above Task:

- Difference between primitive and non-primitive types?
- What is type casting?
- Why is Scanner preferred over BufferedReader?
- What is default value of variables?
- Explain scope of variables.

📌 **Task Submission Guidelines**

- ⏰ **Time Window:**

 Youcan complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

 Youare free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🛠️ **Debug Yourself:**

 Tryto resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💸 **No Paid Tools:**

 Ifthe task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.
Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

📤 **Submit Here:**

 After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [Submission Link ]